LECTURE -18

Page 1: Karyotyping (G-banding)

This page introduces the technique of **Karyotyping**, specifically using **G-banding**[111]. It features an image of a complete human karyotype.

Karyotyping Overview

- **Definition:** Karyotyping is a process that visualizes and organizes chromosomes to analyze their number and structure[2].
- **Procedure:**
    1. **Isolation:** Metaphase DNA is isolated from the nuclei of cells[3].
    2. **Staining (G-banding):** The DNA is stained with a dye, which produces a unique pattern of bands on each chromosome, known as **G-banding**[444].
    3. **Imaging:** The stained chromosomes are then lined up on a slide in homologous pairs and photographed/imaged[5].
- **The Karyotype Image:** The image on this page displays the 22 pairs of **autosomes** (numbered 1 through 22) and the **sex chromosomes** (X and Y)[6]. The chromosomes are visible as paired structures (each consisting of two sister chromatids). The banding patterns help in identifying and matching the homologous pairs[7]. This specific karyotype, with one X and one Y chromosome, represents a **male**[8].

Page 2 & 3: Spectral Karyotyping and Structural Abnormalities

These pages introduce a more advanced karyotyping technique and illustrate various structural changes that can occur in chromosomes.

Spectral Karyotyping (SKY)

- **Concept:** Spectral karyotyping is another method for chromosome analysis[9].

- **Advantage:** This technique makes it **visually easy to identify chromosomal structural abnormalities**[10]. Unlike G-banding, which relies on light and dark patterns, SKY uses multiple fluorescent dyes to "paint" each chromosome pair a different color[11]. This is visualized in the image on page 3 .

Consequences of Chromosomal Breakage

- **Gene Changes:** When chromosomes break and rejoin in an abnormal way (structural abnormality), genes can also **"break"** and **"fuse"**[12].
- **Functional Effects:** This can lead to a **loss of function** for the original gene or the creation of **"fusion genes"** which have an **abnormal function**[13].

Page 4 & 5: Structural Abnormalities and the Philadelphia Chromosome

Page 4 illustrates several types of structural abnormalities, and Page 5 details a specific, clinically relevant translocation: the Philadelphia chromosome.

Types of Structural Abnormalities

The figure on page 4 illustrates different ways a chromosome's structure can be altered[14]:

- **Deletion:** A segment of the chromosome is **removed**[15].
- **Duplication:** A segment of the chromosome is **repeated**, resulting in a longer chromosome[16].
- **Insertion:** A segment from one chromosome is **added** to a nonhomologous chromosome[17].
- **Reciprocal translocation:** Nonhomologous chromosomes **exchange segments**[18].
- **Robertsonian translocation:** Translocation that involves fusion of two acrocentric chromosomes near the centromere with a loss of the short arms[19].
- **Inversion:** A segment of a chromosome is **reversed** in orientation. This can be **paracentric** (not including the centromere) or **pericentric** (including the centromere)[20].
- **Y chromosome isodicentric:** A structurally abnormal Y chromosome that is identical on both sides of the centromere[21212121].

- **Ring or Marker chromosome:** A chromosome that forms a **ring structure** after both ends are deleted and the sticky ends join[22222222].

The Philadelphia Chromosome

This is a classic example of a reciprocal translocation[23].

- **The Break:** It results from a **translocation between chromosomes 9 and 22**[24].
- **The Resulting Chromosomes:**
  - It creates a **longer chromosome 9**[25].
  - It creates a **changed chromosome 22**, which is called the **Philadelphia chromosome**[26].
- **The Fusion Gene:** The translocation moves the **abl gene** (from chromosome 9) next to the **bcr gene** (on chromosome 22), creating a **$bcr-abl$ fusion gene** on chromosome 22[27].
- **Oncogene Function:** The **$bcr-abl$** fusion gene acts as an **oncogene**[28]. An oncogene is a gene whose activity causes cancer[29]. This specific fusion is associated with chronic myeloid leukemia (CML) .

Page 6 & 7: Gregor Mendel and Pea Plants

These pages transition to genetics, focusing on Gregor Mendel's experimental system—the pea plant.

Pea Plant Reproductive Structures

- **Male:** The male structure is the **stamen**, composed of the **anther** (produces pollen) and the **filament** (stalk)[30303030].
- **Female:** The female structure is the **carpel**, composed of the **stigma** (receives pollen), the **style** (stalk), and the **ovary** (contains the **ovules**)[31313131].

Mendel's Experimental Mating

- **Fertilization:** Pea plants can naturally **self-fertilize** or **cross-fertilize**[32].

- **Controlled Crossing:** Mendel's key technical advantage for experimental mating was the ability to perform controlled cross-pollination (crossing)[33].
  1. He could take **pollen** from the **anther** of one flower and **dust it onto the stigma** of another flower[34].
- **The Crossing Procedure (Figure 14.2 of Campbell's Biology):**
  1. **Preparation:** To prevent self-pollination, Mendel removed the **stamens** (male parts) from the purple flower[35].
  2. **Pollination (Crossing):** He transferred sperm-bearing **pollen** from the white flower's stamens to the egg-bearing **carpel** of the purple flower[363636363636363636363636363636].
  3. **Maturation:** He waited for the pollinated carpel to mature into a **pod**[37].
  4. **Planting:** The seeds from the pod were planted[38].
  5. **F_1 Generation Result:** All the offspring (the **First filial generation, F_1**) had **purple flowers**[39393939].
  6. **F_2 Generation:** The F_1 hybrids were allowed to **self- or cross-fertilize**[40].
  7. **F_2 Generation Result:** The **Second filial generation (F_2)** produced **705 purple-flowered plants** and **224 white-flowered plants**[41].
- **Reciprocal Cross:** The same results were obtained when the cross was reversed (**reciprocal cross**), transferring pollen from purple flowers to white flowers[42].

Page 8, 9, 10 & 11: Law of Segregation

These pages delve into Mendel's experimental design and his first principle of heredity.

Mendel's Core Concepts

- **Characters/Traits:** Mendel chose to study characters that occurred in two distinct, alternative forms (e.g., purple or white flower color)[43].
- **True-breeding:** He started with varieties that, over many generations of self-pollination, produced only the same variety as the parent plant[44].
  - The **true-breeding parents** are the **P generation** (parental generation)[45454545].

- **Hybridization:** The mating of two true-breeding varieties is called hybridization[46].
  - The hybrid offspring are the **F_1 generation** (first filial generation)[47474747].
  - The offspring of the F_1 generation are the **F_2 generation** (second filial generation)[48484848].

Observations Leading to the Law of Segregation

- **F_1 Generation:** When true-breeding purple (P) flowers were crossed with true-breeding white (P) flowers, all F_1 plants had **purple flowers**[49494949]. The information for white flowers seemed to disappear[50].
- **F_2 Generation:** When the F_1 hybrids self- or cross-pollinated, the F_2 generation saw the **reappearance of white flowers**[51515151].
  - The traits appeared in a quantitative **ratio of 3:1** (3 purple-flowered plants : 1 white-flowered plant)[52].

Dominant and Recessive Traits

- Mendel reasoned that the factor for white flowers was **hidden, or masked**, when the purple-flower factor was present in the F_1 plants[53535353].
- **Dominant Trait:** The trait that determines the organism's appearance (e.g., purple flower color)[54545454].
- **Recessive Trait:** The trait that has no noticeable effect on the organism's appearance when the dominant allele is present (e.g., white flower color)[55555555].

Mendel's Model and the Law of Segregation

Mendel's observations were explained by a four-part model:

1. **Alternative Versions of Genes (Alleles):** Variations in inherited characters are accounted for by alternative versions of genes, called **alleles**[56].
2. **Two Alleles per Character:** For each character, an organism inherits two copies (two alleles) of a gene, one from each parent[57].

3. **Dominance:** If the two alleles at a locus differ, the **dominant allele** determines the organism's appearance, and the **recessive allele** has no noticeable effect[58].
4. **Law of Segregation:** The two alleles for a heritable character **segregate (separate from each other) during gamete formation** and end up in different gametes[59].

The table on page 11 shows that Mendel observed this characteristic **~3:1 ratio** across **seven different characters** in pea plants[60].

Page 12 & 13: Alleles, Chromosomes, and the Punnett Square

These pages connect Mendel's abstract "heritable factors" to physical alleles and show the mechanism of segregation and fertilization using a Punnett Square.

Alleles and Chromosomes

- **Alleles:** Alleles are **alternative versions of a gene**[61].
- **Locus:** The specific position on a chromosome where a gene for a character (like flower color) is found is called the **locus**[62].
- **Diploid Cells:** In a diploid cell at metaphase, there are **4 alleles of a gene** (one on each sister chromatid of the homologous pair)[636363636363636363].
- **Gametes:** A haploid gamete has **one chromatid** at the end of meiosis, meaning it carries **one allele**[64].
- **Molecular Basis of Alleles:**
    - The **purple-flower allele** may result from a specific DNA sequence that leads to the production of an **enzyme** that synthesizes purple pigment[6565656565656565656565656565].
    - The **white-flower allele** may have a different DNA sequence (a mutation) that results in the **absence of that enzyme**, thus producing no pigment[666666666666666666].

The Law of Segregation Illustrated (Punnett Square)

This section uses a Punnett Square to visualize the cross, using P for the dominant (purple) allele and p for the recessive (white) allele.

- **P Generation (True-breeding):**
  - **Purple Flower:** Genetic makeup **PP** (diploid, two copies of the dominant allele)[6767676767].
  - **White Flower:** Genetic makeup **pp** (diploid, two copies of the recessive allele)[6868686868686868].
  - **Gametes:** The P plants produce only one type of haploid gamete: **P** or **p**[69696969].
- **F_1 Generation:**
  - **Fertilization:** When P gametes fuse ($\text{P} + p$), they form the diploid combination **Pp**[70].
  - **Appearance/Genetic Makeup:** The plant has a **Purple flower** and a genetic makeup of **Pp** (hybrid), because P is dominant over p[717171717171717171].
  - **Gametes (Segregation):** Each F_1 plant produces two types of gametes, $\frac{1}{2}$ **P** and $\frac{1}{2}$ **p**, which illustrates the **law of segregation**[72727272].
- **F_2 Generation:**
  - **Fertilization:** F_1 gametes are combined (self or cross-fertilization) using the Punnett Square[737373737373737373].
  - **Diploid Combinations (Genotypes):** Three diploid combinations are possible[74]:
    - $\frac{1}{4}$ **PP** (Purple) [75757575]
    - $\frac{1}{2}$ **Pp** (Purple) [767676767676767676]
    - $\frac{1}{4}$ **pp** (White) [77777777]
  - **Phenotypic Ratio:** The result is a **3:1 phenotypic ratio** of Purple flowers to White flowers[78]. The recessive trait (pp, White) reappears after skipping one generation[79].